Jiyang Lee

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Research Interests

Robot Learning, Generative Robot Policies, Vision-Language-Action Models, Humanoid Loco-Manipulation

Research Experience

Seoul National University, Seoul, Korea *Mar 2024 – Present*
Graduate Researcher (Advisor: Prof. Seong-Woo Kim)

- Built and evaluated generative robot policies on a real robot manipulator, from teleoperated demonstrations to closed-loop deployment (with Prof. Roya Firoozi, University of Waterloo).
- Developed inference-time guidance for pretrained generative policies using learned visual dynamics.
- Studying whole-body loco-manipulation on a humanoid robot.

Education

Seoul National University, Seoul, Korea *Mar 2024 – Present*
Ph.D. Candidate in Civil and Environmental Engineering
Advisor: Prof. Seong-Woo Kim
Expected graduation: Aug 2028

University of Waterloo, Waterloo, Canada *Jan 2026 – Jul 2026*
International Visiting Graduate Student, Electrical and Computer Engineering
Advisor: Prof. Roya Firoozi

Sogang University, Seoul, Korea *Mar 2015 – Feb 2017*
Master of Science in Mechanical Engineering
CAD/CAM Lab, Advisor: Prof. Cheol-soo Lee

Sogang University, Seoul, Korea *Mar 2011 – Feb 2015*
Bachelor of Science in Mechanical Engineering
Undergraduate Research Assistant, CAD/CAM Lab (Dec 2013 – Feb 2015)

Professional Experience

LG Display Co., Ltd., Paju, Korea *Dec 2020 – Mar 2024*
Research Engineer, Vision Algorithm Team

- Developed a few-shot classification model for OLED defect images.
- Built an AI-based automatic anomaly detection and alarm system.
- Accelerated a legacy rule-based anomaly detection system 7× with CUDA.

Korea Institute of Science and Technology (KIST), Seoul, Korea *Oct 2018 – Feb 2020*
Robotics Researcher

- Designed and assembled a quadruped robot.
- Developed a panoramic-view system for a firefighting armored robot.
- Conducted UWB-based robot swarm localization experiments.

CSCAM Co., Ltd., Gwangju, Korea *Jan 2017 – Sep 2018*
CNC Department

- Built vision and XRF inspection machines and OLED line loading equipment.

Publications

Under Review

- **J. Lee**, R. Firoozi, and S.-W. Kim, title withheld for double-blind review, submitted to IEEE ICRA 2027.

Conference Papers

- C. Kim, K. Kim, M. Oh, H. Baek, **J. Lee**, et al., "E2Map: Experience-and-Emotion Map for Self-Reflective Robot Navigation with Language Models," *IEEE ICRA*, 2025.
- M. Kim, **J. Lee**, et al., "A Robust Position Estimation Algorithm under Unusual Large Range Errors," *IEEE RO-MAN*, 2019.
- **J. Lee** and C.-S. Lee, "Path Planning for SCARA Robot Based on Marker Detection Using Feature Extraction, Labeling, and Inverse Perspective Transform," *FAIM*, 2016.

Journal Articles

- **J. Lee** and C.-S. Lee, "Path Planning for SCARA Robot Based on Marker Detection Using Feature Extraction and Labelling," *International Journal of Computer Integrated Manufacturing*, 31(8):769–776, 2018.
- **J. Lee**, D.-G. Lee, and C.-S. Lee, "Development of Robotic Dispensing System for Radiopharmaceuticals Using SCARA Robot," *Journal of the Korean Society of Manufacturing Technology Engineers*, 26(5):441–449, 2017.

Awards

- Best Paper Award, Korean Society of Manufacturing Technology Engineers, 2017

Leadership and Service

Conference Reviewer *2026 – Present*
IEEE IROS 2026, IEEE ICRA 2027

SNU SOLO Solar Car Team, Seoul National University *2025*
Advisory Leader, 2025 Bridgestone World Solar Challenge, Australia